

Towards fluency-structured dialogues with natural speech input

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This paper presents the results of a study into how structured dialogue techniques in IVR systems can be extended to allow more natural speech interaction with the caller. A spoken language system was produced which allows callers to set reminder calls or bar outgoing or incoming telephone calls. In spite of the limits of speech recognition performance, the resulting system had a highly natural speech interface which allowed the caller to optionally offer one or more pieces of information at a time.

1. Introduction

Isolated word speech recognition over the telephone network has now become accurate enough for many types of large vocabulary interactive voice response (IVR) tasks [1, 2]. These systems are typically characterised by structured speech dialogues using pre-recorded service prompts. Typical dialogues follow a menu structure, e.g. 'Please say marketing, sales or customer service,' or form filling, e.g. 'Please say the surname.. Attwater.. Now please say the firstname.' In each of these cases, the speech recogniser at a given node in the dialogue is restricted to one of a set of currently available words.

One of the limitations of this approach is that quite verbose dialogues are needed as callers can rarely be expected to read priming material. Such dialogues quickly become annoying to regular users and this has led to the need for techniques such as fast-track dialogues and cut-through.

This paper considers an alternative approach to this problem — a structured dialogue with flexible entry of information using recognition of fluent speech. The key characteristic is the use of short, open questions using speech recognition grammars and flexible dialogue structures to manage the variability in response. The potential benefit of this approach is that it allows callers to use sentences and to offer information in a flexible way.

The domain chosen by the study was automation of two of the BT Select Services — *Call Barring* and *Reminder Calls*. Call Barring enables customers to selectively bar different types of outgoing calls and also to bar all incoming calls, if desired. The Reminder Call service enables customers to book calls from the telephone exchange at a

specified time, either as a single alarm, or at regular intervals. The study resulted in an experimental service being built which was named *Freedom*.

Although full spoken language systems are under investigation at BT Laboratories (BTL) [3], Freedom deliberately kept many of the restrictions of structured dialogue paradigms. This was in order to explore the extent to which the existing frameworks of structured dialogues could be extended to incorporate elements of natural language. In order to study this, the following restrictions were deliberately retained:

- no user initiatives were allowed,
- callers had to answer the question asked,
- only mixtures of pre-recorded speech prompts were used for speech output,
- a structured dialogue service creation tool, BT Visage [4], was used for the dialogue implementation.

2. System architecture

Figure 1 shows a simplified system architecture for Freedom. In addition to the usual components of an IVR system, namely recognition, speech output and a dialogue controller, an additional layer comprising a parser, blackboard, and prompt generation were found to be necessary. These or similar components are to be found in most spoken language systems, but in this system, these components were deliberately kept very simple.

Figure 2 shows the blackboard in more detail. It contains three logical stores of information. The first, task feature-values, keeps a cumulative record of all of the

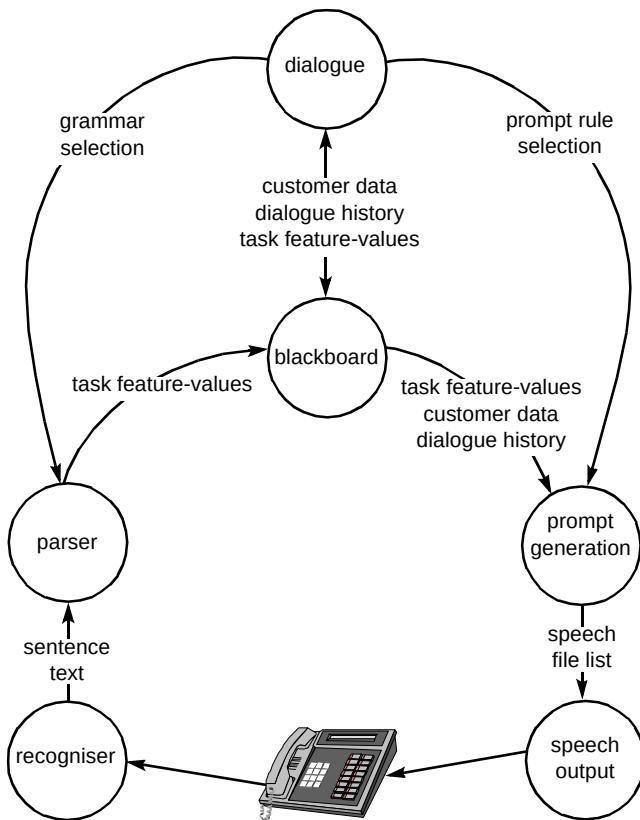


Fig 1 Simplified system architecture.

information entered by the caller while attempting a current task. Only one task may be worked on at once and it is cleared each time a task is completed or aborted. The second, dialogue history records key dialogue events in the current call. This helps to keep continuity in the dialogue allowing prompt generation to take into account what the caller has heard in previous prompts when formulating the current prompt. Finally, customer data is a persistent store of user settings. It contains, for example, current reminder call and call barring settings, and the user password.

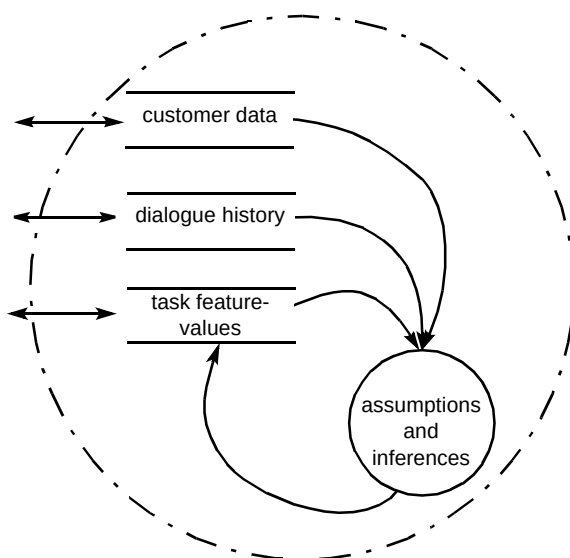


Fig 2 The blackboard in more detail.

It can be seen in Fig 2 that the task feature-values are modified by assumptions and inferences. These may be viewed as rules that are continuously monitoring the state of the blackboard to see whether additional feature-values may be inferred or assumed given the current blackboard state. This will be explored in greater detail later.

3. Grammars and parsing

Figure 3 shows the process of recognition and parsing in more detail. For each question node, a recognition grammar was defined which describes the expected responses to a particular question. The recogniser then returns a candidate phrase. The parser then attempts to make sense of this sentence, returning values for any task feature-values that have been mentioned in the response. Put simply, the recogniser returns what was **said**, and the parser returns what this **means**.

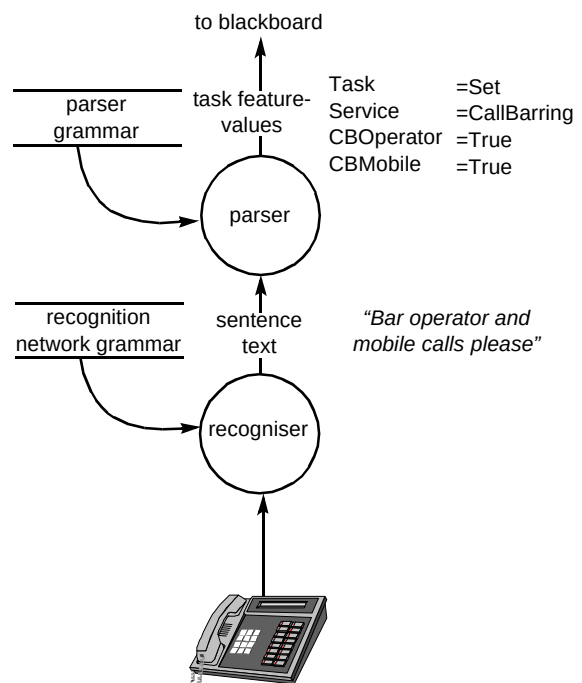


Fig 3 Recognition and parsing.

In the Freedom system, the recognition network grammar and the parser grammar were derived from a single definition. This meant that the recogniser could not return sentences that the parser did not know how to parse.

3.1 Grammar design

The grammar designs in Freedom were based on a finite state model of allowable sequences of words and phrases. With this technique, the exact wording of responses must be anticipated. Figure 4 shows an example of a simple finite state grammar modelling the responses expected from the question: 'Would you like the regular reminder call daily, weekly or on weekdays?'

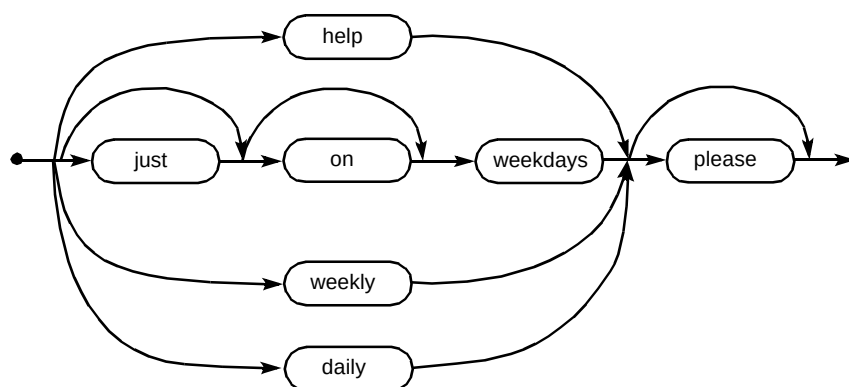


Fig 4 A simple finite state grammar example.

Grammars in the system for responses to questions such as: 'How may I help you?' were considerably more complex than the example. It was the flexibility of these grammars, allowing multiple features to be offered in a single sentence that made Freedom different from ordinary structured dialogue systems.

Because callers may say sentences outside of the scope of the grammar, this approach to grammar design is known to be brittle. It is, however, more accurate than other schemes if the language of the caller matches the grammar. It is possible that word sink models, which are general models representing words that are not interesting to the system, could be mixed into the grammar to make it less brittle [5].

The initial grammar designs were produced manually by considering the exact wording of the question prompt, and the features that were expected to be elicited by this question. From this a draft grammar was built which was then tested using small-scale user trials observing real-life behaviour in the dialogue. It is believed to be necessary to go through several iterations of such trials before fully representative grammars are obtained.

4. Dialogue design

4.1 Dialogue design principles

The purpose of this study was to test whether a certain style of dialogue interaction could be built and whether it would be usable. As a starting point, the following dialogue design principles were chosen:

- no focus shifting is supported — callers must complete one task at a time,
- no priming should be necessary before a caller can use the system,
- context-sensitive help should be available at every recognition node,
- all changes to the customer data should be explicitly actioned with a closed question,

- due to the short nature of most tasks, they may only be aborted by hanging up and starting again,
- confirmation by the caller may include extra information, e.g. 'Yes, I'd like to bar operator calls please,'
- although not excluded by the architecture, spontaneous contradiction by the caller is not currently supported except at explicit yes/no questions, e.g. 'No, it should have been 12.30,' is not supported,
- in line with existing dialogue practice, three-level prompting should be employed at every recognition node (see section 4.3),
- second- and third-level prompts will repeat questions in an increasingly explicit way, and may offer isolated word options, but the speech recognition grammars will remain constant for all levels.

These dialogue principles were associated with a number of architectural principles:

- all system prompts will use pre-recorded speech (no text-to-speech synthesis),
- each question node has its own pre-built speech recognition grammar,
- once a feature-value has been entered by the caller the dialogue will not use a subsequent grammar which contains that feature,
- only the top sentence candidate will be considered by the parser.

4.2 Dialogue structure representation

Every dialogue design must establish a structure which models the interaction between the human and the computer. Most structured dialogue service-creation tools, including BT's Visage, which was used to implement the Freedom system, represent dialogue structures graphically using dialogue nodes, and transitions between these nodes when certain conditions are met.

In Freedom, the structure of the dialogue is designed around a number of question or information nodes. Each of these nodes represents a single logical question, e.g. 'At what time?' or a single logical statement, e.g. 'Confirmed, reminder call set at 12.13 p.m.' Transition between these nodes is based upon the state of the blackboard. This view of dialogue structure fits in well with current structured dialogue paradigms. The difference with Freedom, however, is that the whole blackboard is used and the flexible speech input means that there are many more potential transition conditions to consider than in an ordinary isolated word dialogue. This introduces a problem of design complexity which must be managed.

Figure 5 shows a simplified view of the different components from which the Freedom dialogue was hierarchically composed. At the core of the dialogue are three-level question nodes. Each node is configured to ask one logical question. Target blocks then sequence the use of these question nodes and also simple information nodes (not shown in the diagram for simplicity) to ensure that particular target criteria are satisfied. The dialogue itself then sequences the use of these target blocks to ensure that the correct service dialogue is achieved. This section describes these different components in more detail.

4.3 Three-level question nodes

Figure 6 shows the structure of the three-level question nodes in more detail. At the start of each node, a grammar is selected and the prompts are generated, taking into account the state of the blackboard.

The corresponding recognition grammar is then used in a three-level prompting structure. If the recogniser accepts a given input, then the phrase is parsed. As discussed in the previous section, any permutation of a number of features may be returned by the parser. These features are then added to the blackboard. Any new inferences or assumptions which can be made using this new information will then take effect and the node is complete.

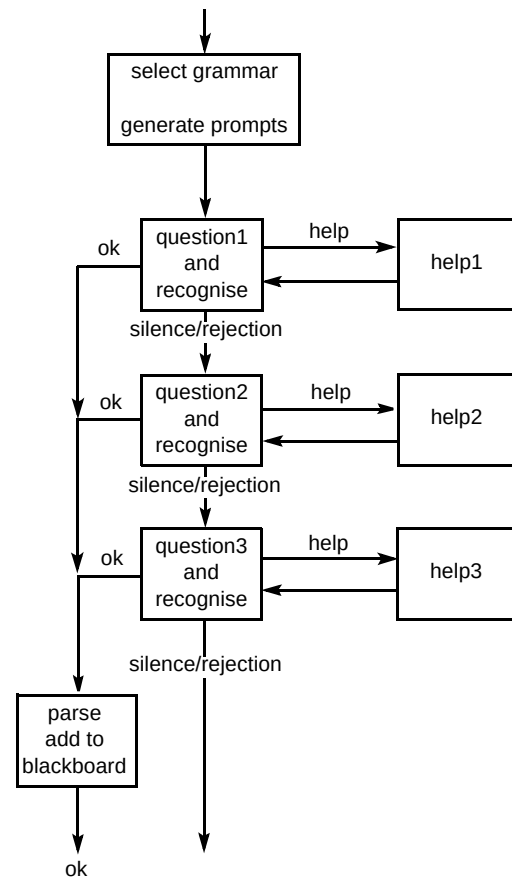


Fig 6 Three-level question node.

If silence is heard or the recogniser is uncertain then the next-level question is asked. This is a repeat of the same question but with increasingly explicit wording. If 'Help' is heard at any level then an appropriate help prompt is played out and the question is repeated. Only three levels are permitted. If there are more than three successive rejections or silences then the dialogue will politely terminate.

4.3.1 Target blocks

Figure 7 shows a portion of structured dialogue from the implemented service. It shows two three-level question nodes. The first contains a very open question — 'How may

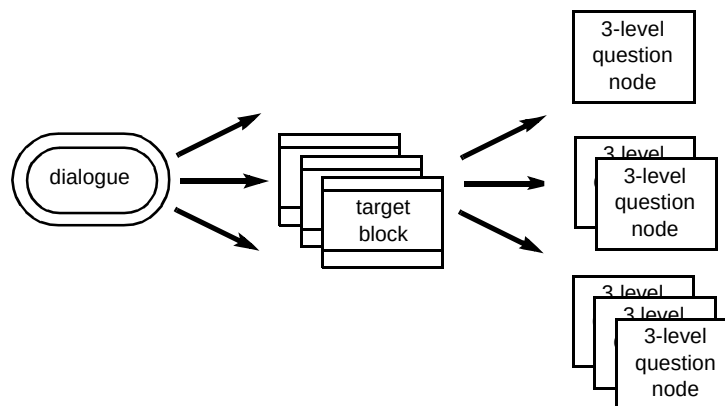


Fig 5 A simplified view of how the dialogue structure was hierarchically designed.

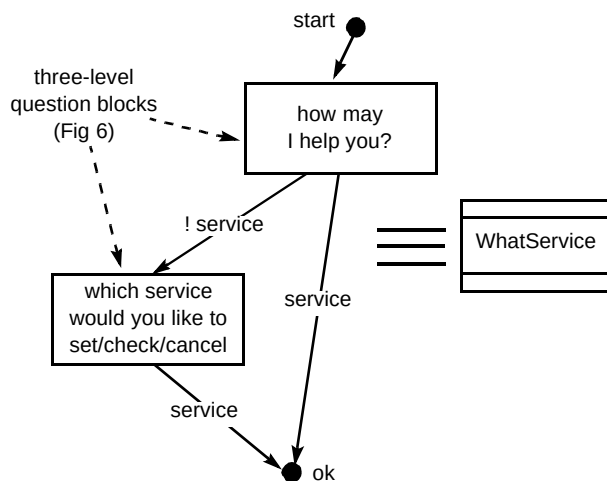


Fig 7 The structure of question nodes in the 'WhatService' target block.

I help you?' — and has a similarly open grammar to deal with the response. Figure 3 shows an example of the sort of response this question may receive and the corresponding features returned by the parser and placed on the blackboard.

The blackboard is then tested to see whether the feature 'Service' has been instantiated yet or not. If it has, then this

portion of the dialogue completes. If not, then the dialogue moves to the second question node. This asks the more closed question 'Which service would you like?' (The exact wording of this prompt may be modified by some of the information that was offered to the previous question — this will be discussed later.) The grammar for this node is restricted to recognise just the service names available in the application, in this case: 'Reminder Call', and 'Call Barring' or any other alias for these services. Thus the parsed output of this recognition must result in the Service feature being set on the blackboard.

This portion of dialogue therefore asks a very open question, but then uses a further closed question if the service name was not offered (or inferred) in the initial response. Thus the net effect of this dialogue portion is to guarantee that the service feature is set when this block completes. This is an example of a target block, which is a very powerful tool in helping the dialogue designer divide and conquer the interaction into manageable chunks.

4.3.2 Dialogue structure and fall-through

Figure 8 shows how target blocks are used in the outline service structure for setting reminder calls as part of the service dialogue. The interaction is broken down into a

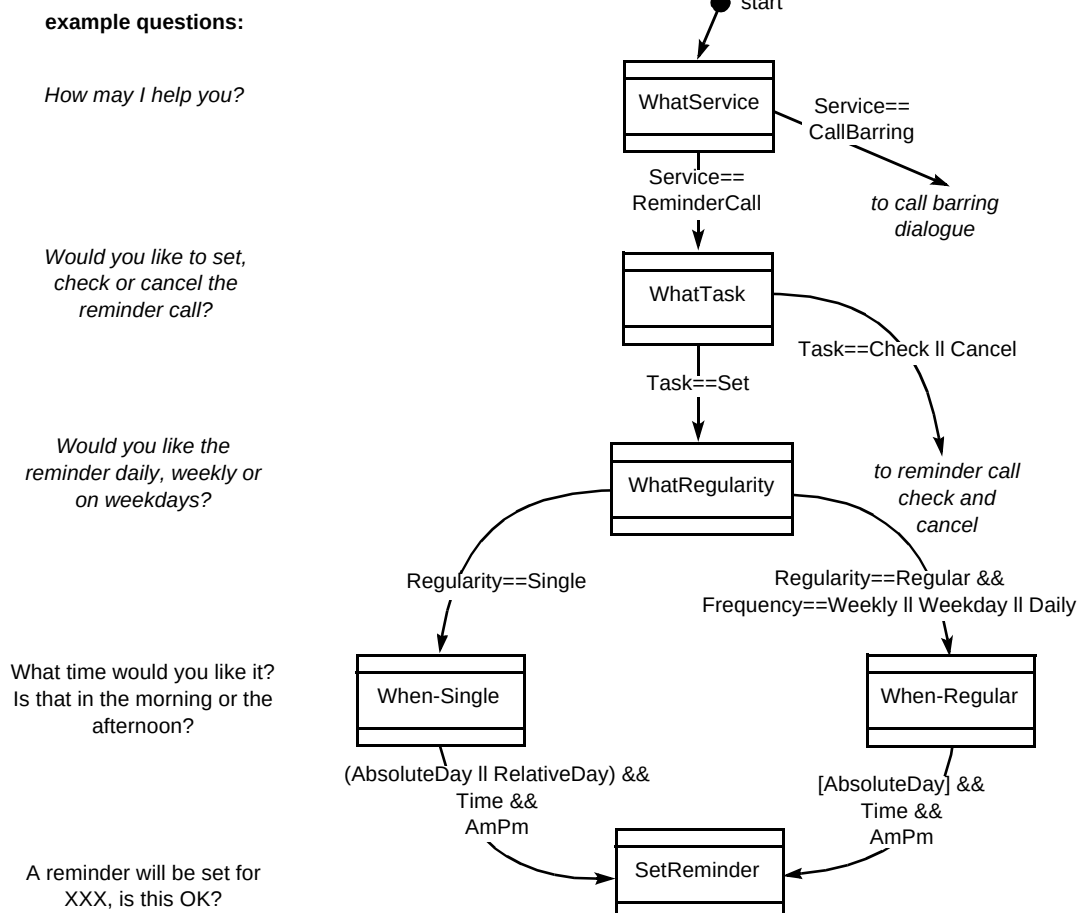


Fig 8 The portion of the outline dialogue structure used for setting reminder calls.

number of successive targets which must be satisfied in sequence in order to complete the current task.

The WhatService target block of Fig 7 can be seen at the top of Fig 8. Each block in the figure has target completion criteria similar to the WhatService block and contains structured dialogues comprising information and question nodes designed to ensure that this completion criteria is satisfied before the block can complete. These completion criteria are based on values on the blackboard and are shown on the transitional arcs out of the block. Where there are a number of transitional arcs, they represent one of a number of possible completion criteria for that particular block.

The structure shown in Fig 8 is very similar to that of an ordinary structured dialogue with one key difference. In ordinary structured dialogues, a given block is designed to satisfy only the target of the current block. With the use of open questions and speech recognition grammars that may cause multiple features to be returned by the parser, the criteria for a later target block may be satisfied by a response to a question in an earlier block. In these cases, the later block simply completes immediately without entering into any interaction with the caller. This mechanism has been termed 'fall-through' by the authors.

By way of an example, the first question node in the WhatService block asks: 'How may I help you?' and may receive the response: 'I would like to set up a daily reminder call please.' The parsed features of this sentence return the service name (reminder call) the task (set) and the regularity (daily). In this case the dialogue flow will fall straight through the WhatTask block and the WhatRegularity block because their corresponding completion criteria are already satisfied, and enter the When-Regular block which will begin to satisfy its criteria by asking the question: 'On which day of the week would you like the daily reminder call?' Once all of the necessary details for setting the reminder call are completed, the When-Regular block will complete and the SetReminder block will confirm and set up the new reminder call. The interaction is then complete.

5. Prompt generation

Sound dialogue design requires that the wording and intonation of each information or question prompt is carefully designed to take into account the prompts and spoken inputs which have preceded it. In the case of a system that uses only pre-recorded speech but allows more fluent speech input, this presents a particular problem. The specific wording of prompts in this case depends on the state of the blackboard. Figure 9 shows the architecture used to support rule-based generation of prompts.

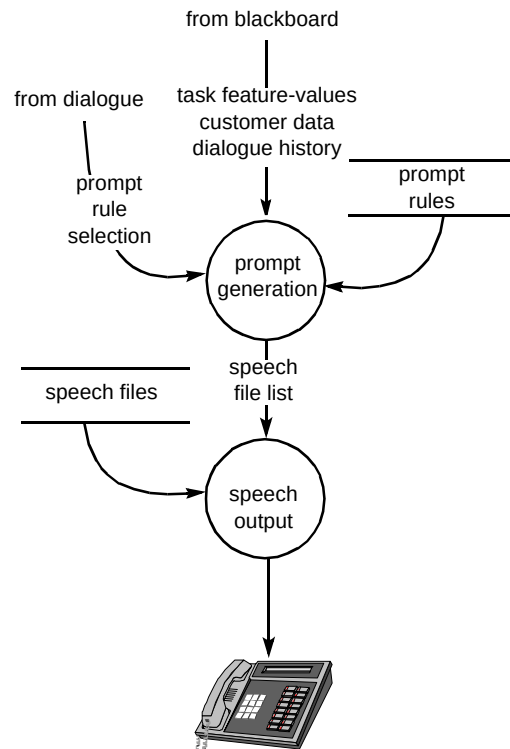


Fig 9 Prompt generation in more detail.

Figure 10 shows an example dialogue fragment from Freedom which will be used to illustrate some of the issues surrounding prompt generation.

How may I help you?
Can I have a reminder call on Tuesday please?
Set Reminder Call, thankyou. You currently have a single reminder call booked at five thirty on Thursday morning. Would you like to replace it?
Yes please
What time on Tuesday would you like the reminder call?
Three thirty
Is that in the morning or the afternoon?
The morning.
A single reminder call will be booked at three thirty on Tuesday morning. Is this OK?
Yes.
Confirmed. Reminder call booked...

Fig 10 Example dialogue fragment.

5.1 Dialogue continuity

In ordinary structured dialogues, a given prompt will usually be preceded by one of a few pre-determined prompts. In more fluent systems, however, the effect of fall-through will mean that many information or question nodes may have been preceded by any one of a number of nodes. This presents a continuity problem to the dialogue designer.

In addition to this problem it is helpful to use references word such as 'it' and 'them' in the output dialogue, e.g. 'Is

that in the morning or the afternoon?’ These are known as anaphora. This reduces the tedium of constant repetition of key phrases, e.g. ‘What time would like the reminder call?’ followed by: ‘Would you like the reminder call in the morning or the afternoon?’ The danger of using these anaphoric references in the output within a fall-through environment is that the ‘it’ to which the prompt is referring may become unclear. This is why dialogue history (nodes visited) and task feature-values (features that have been input in the current enquiry) are both required during prompt generation.

5.2 Embedded confirmation

Natural human/human dialogues are full of subtle confirmations from the listener that the talker has been heard and understood. This is also important in automated dialogues. Traditional structured dialogues often use explicit confirmation — ‘Did you say Edinburgh’ — or implicit confirmation — ‘Edinburgh, thankyou’ — to perform this function.

In order to establish a natural feel and to cope with the effects of fall-through, the dialogue in Freedom experimented with implicit confirmations that were intimately bound up in the language of the next question or statement — termed ‘embedded confirmations’.

Consider the dialogue in Fig 10. The caller asks for a reminder call on Tuesday in their reply to the first question. The system then warns them that they already have a reminder call booked and confirms that they would like to replace it. Traditional implicit confirmation of the task is used here — ‘Set Reminder Call, thankyou.’ The following question then turns the subject to time entry (since the task and service target blocks are both satisfied) with the question — ‘What time on Tuesday would you like the reminder call?’ This embedded confirmation of the day reassures the caller that the initial input of ‘Tuesday’ was not lost and gives a very fluid feel to the dialogue. Following the same pattern, the following question could ask: ‘Is that three thirty in the morning or the afternoon?’ but it was not felt necessary in this case.

In reality, the design complexity of these forms of confirmation can rapidly explode and the dialogue designer must balance the competing requirements for simplicity against the need for clarity within the dialogue design.

6. Inferences and assumptions

Blackboard-based inferences and assumptions were found to be vital in the Freedom system in order for it to maintain a usable dialogue and avoid redundant or annoying questions to the caller.

6.1 Inferences

Inference rules were used to relate feature-values which may be deduced without actually being mentioned by caller.

Their role is to eliminate redundant questions based on simple logical relationships.

Figure 11 shows how features were found to be related in the analysis of Freedom. Each feature is shown as a box containing smaller boxes representing the mutually exclusive values for the feature.

Inference rules may be derived from this representation. Consider the case of a caller asking: ‘Can I have a weekly reminder call please?’ in response to the first question. The feature Regularity will be set to ‘Weekly’, Task to ‘Set’ and Service to ‘ReminderCall’. If a weekly reminder call is being requested then, by inference, it must be a regular reminder call and Frequency may then be inferred to be ‘Regular’.

Another example of the use of inferences is for time entry. An afternoon caller requesting a reminder call at ‘four o’clock today’ must be referring to four o’clock that afternoon and the feature AmPm may then be inferred to be ‘p.m.’

6.2 Assumptions

Assumptions are similar to inferences. They provide a way of offering callers default behaviour resulting in faster interactions for the majority of callers. Figure 12 shows an assumption mechanism at work in the Select Services Reminder Call dialogue. In the example, the caller is assumed to be wanting a reminder call today because they have only offered an unqualified time. The initial user analysis indicated that this assumption would suit most callers.

7. Speech recognition

7.1 Recognition performance

Freedom used the BT STAP speech recogniser for its development. This is a speaker-independent large-vocabulary telephony recogniser enabling finite state grammars to be built using British English phoneme recognition models.

Flexible large-vocabulary speech recognition over the telephone is still a considerable challenge and the performance of speech recognition is still the limiting factor in many speech systems to date. This was also found to be true in Freedom.

One of the factors found to significantly affect recognition performance was the pronunciation used for the word models. Many pronunciation lexicons quote only a single preferred form for the word. It has been observed, however, that creating multiple pronunciations for words, which account for individual variations in accent and pronunciation, can significantly improve the recognition performance.

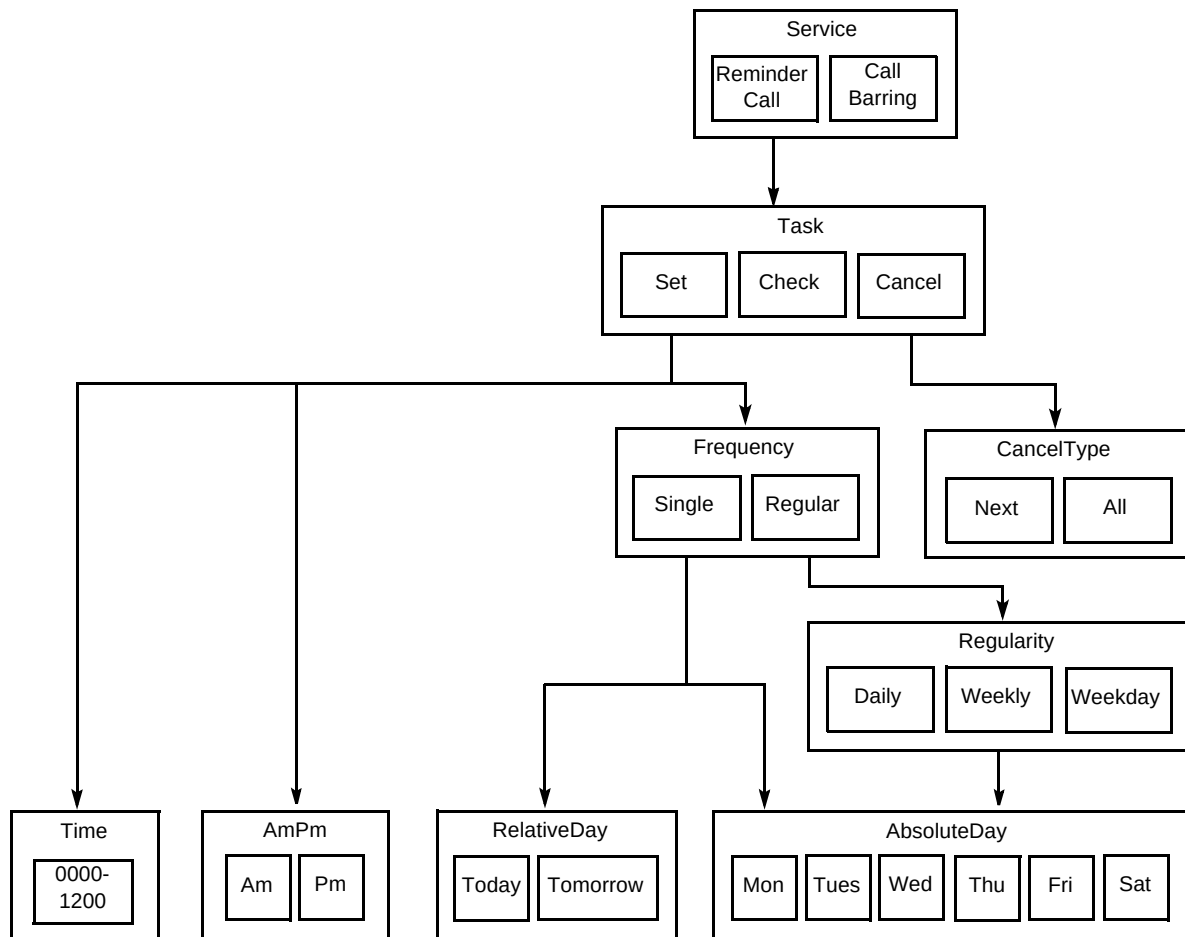


Fig 11 Feature inference diagram for reminder calls.

How may I help you?
Can I have a single reminder call please

When would you like the reminder call?
Three thirty

A single reminder call will be booked at three thirty this afternoon.
 Is this OK?
Yes

Confirmed. Reminder call booked.
 How else may I help you?

Fig 12 Assumptions at work in the Select Services Reminder Call dialogue.

8. User interface trials

The user interface developed for Freedom is experimental. The assumptions upon which it is built are currently being verified in user interface trials. Topics for study will include the following.

- Habituation
 How quickly does the dialogue educate callers to use preferred vocabulary?

- Longitudinal Close-down
 Do long-term users close their behaviour down to isolated word responses in order to take advantage of enhanced recognition performance?
- First prompt
 How important is the wording of the first question in establishing the style of responses expected?
- Contradictions
 Do callers spontaneously try to repair errors as they occur, or do they clear down and try again?
- Priming
 Can callers use this style of dialogue with minimal priming?

9. Conclusions

The Freedom system has established that it is possible to implement simple spoken language systems using established graphical structured design tools. Many of the design techniques for structured dialogues with natural speech input still appear to be appropriate for simple, well-scoped, tasks but some additional tools and frameworks are

needed to keep the design complexity manageable. Techniques identified in this study include target blocks, fall-through, question close-down, and frameworks for inferences, assumptions, and prompt generation. The benefits of allowing natural speech input are that task completion times should be significantly reduced due to the callers ability to enter multiple task feature-values in one utterance.

Recognition accuracy still fundamentally limits what may be achieved, but simpler tasks with well-defined structures and small vocabularies are beginning to show usable recognition performance levels. Usability trials are necessary before the assumptions underlying this trial system are proved, but the Select Services trial system appears to conduct highly intuitive and flexible interactions with dialogue that is much closer to normal human/human styles of interaction.

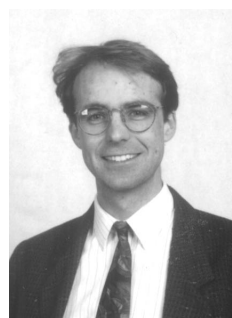
The evolutionary approach taken by this study, starting from current structured dialogue techniques and tools, means that systems such as this could be commercially viable in a very short space of time.

Acknowledgements

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David Attwater received an MEng in electronic systems engineering at the University of York in 1991.

Since then he has worked at BT Laboratories researching telephone applications of speech recognition. Over this period he has developed techniques for accessing large databases using speech, and investigated practical methods for designing structured dialogues with natural speech input. He now leads a team looking at the use of speech technology for robust natural language understanding on telephone helpdesks.



John Fisher received a BA in Engineering/Electrical Sciences from Pembroke College, Cambridge in 1971. He has worked at BT Laboratories on many systems including System X announcements, data and modem product design, data communications network management systems, and more recently the development and support of Callminder.

He contributed to the design of the control aspects of natural speech within structured telephone dialogues. He is currently applying advanced speech recognition techniques to call centre solutions for trial and roll-out applications.



Hilary Greenhow received a BSc in Electronic Engineering from Queen Mary College, University of London in 1975. Since then he has worked on many projects at BT Laboratories. Initially he supported, maintained and wrote software associated with the manufacture of high reliability transistors for use in undersea repeaters. This was followed by a similar role in custom integrated circuit design. In the late 1980s he was involved in the porting of a 'secure Unix' system for BT's M6000 computers. He also contributed to the development of the BT ISAP speech platform. More recently he has been working

with fluent speech and address recognition.